

Question_7:

Question description

You are given an array A of N integers. Now, two functions $F(X)$ and $G(X)$ are defined:

$F(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] < A[Z]$

$G(X)$: This is the smallest number Z such that $X < Z \leq N$ and $A[X] > A[Z]$

Now, you need to find for each index i of this array $G(F(i))$, where $1 \leq i \leq N$. If such a number does not exist, for particular index i , output -1 as its answer. If such a number does exist, output $A[G(F(i))]$

Constraints:

$$1 \leq N \leq 3000$$

$$0 \leq A[i] \leq 10^{18}$$

Input :

The first line contains a single integer N denoting the size of array A . Each of the next N lines contains a single integer, where the integer on the i^{th} line denotes $A[i]$.

Output :

Print N space-separated integers on a single line, where the i^{th} integer denotes $A[G(F(i))]$ or -1 , if $G(F(i))$ does not exist.

✓ Logical Test Cases

Test Case 1

INPUT (STDIN)

8
3 7 1 7 8 4 5 2

EXPECTED OUTPUT

1 4 4 4 -1 2 -1 -1

Test Case 2

INPUT (STDIN)

8
31 71 11 17 81 14 51 21

EXPECTED OUTPUT

11 14 14 14 -1 21 -1 -1

Code:

```
1  /*
2  Question_7:
3  For each element, find the first greater element to its right,
4  then find the first smaller element after that.
5  Print the final element or -1.
6  */
7
8  #include <iostream>
9  using namespace std;
10
11 int main()
12 {
13     int n;
14     cin >> n;
15
16     long long a[n];
17
18     for(int i = 0; i < n; i++)
19         cin >> a[i];
20     for(int i = 0; i < n; i++)
21     {
22         int f = -1;
23         int g = -1;
24         for(int j = i + 1; j < n; j++)
25         {
26             if(a[j] > a[i])
27             {
28                 f = j;
29                 break;
30             }
31         }
32         if(f != -1)
33         {
34             for(int j = f + 1; j < n; j++)
35             {
36                 if(a[j] < a[f])
37                 {
38                     g = j;
```

```
39         break;
40     }
41 }
42 }
43 if(g == -1)
44     cout << -1;
45 else
46     cout << a[g];
47 if(i < n - 1)
48     cout << " ";
49 }
50 return 0;
51 }
```

Output:

Test Case_1:

```
8
3 7 1 7 8 4 5 2
1 4 4 4 -1 2 -1 -1
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
8
31 71 11 17 81 14 51 21
11 14 14 14 -1 21 -1 -1
PS C:\Users\Prabin Yadav\Docum
```